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Assignment No :- 6.

Title :- Database connectivity.

Objective :- Write a program to implement MySQL / oracle database connectivity with any frontend language to implement database navigation operation (add, delete, edit etc)

Software & Hardware Requirments.

- Frontend : python (IDLE, Eclipse)
- Backend : MySQL Server
- Library used : MySQL - connector, python
- OS : window / Linux.
- Hardware : Standard PC / Laptop

Theory :-
Database connectivity allows communication between an application and a database management system.

- In python, the SQL connector module is used to connect a MySQL database. It provide function to :-

- Established a connection to the MySQL server, execute SQL queries (Insert, update, Delete, Select).

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- fetch and display results.
  - Commit changes and close connection Safely using these methods a menu driven program can perform Database Navigation operations such as adding, modifying, viewing, and deleting records.

Steps:-

1) Import the mySQL connector module
2) Establish a connection using MySQL Connect -or. connect() with host, user, password and database.
3) Create a cursor object to executes SQL Statements.
4) create functions for each operation
   - add_record () → INSERT new data.
   - view_record () → SELECT and display data
   - Search_record () → SELECT specific record.
   - edit_record () → UPDATE existing data.
   - delete_record() → DELETE a record.

5) Create a menu-driven loop to call the respective functions based on user choice

6) Commit changes after every modification and close the connection at the end.

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Conclusion:-

Thus database connectivity between python (frontend / and MySQL) (backend) was successfully implemented. This experiment demonstrates how python can serve as a powerful frontend for performing database operations efficiently and interactively.